

CHE 201 Spring 2020 HW 3

Thanh Nguyen

TOTAL POINTS

95 / 100

QUESTION 1

1 Question 1 15 / 20

- 5 pts Wrong equation in convection
- 5 pts Wrong calculation in convection
- ✓ - 5 pts Wrong equation in radiation
- 5 pts Wrong calculation in radiation
- 0 pts correct

QUESTION 2

2 Question 2 20 / 20

- ✓ - 0 pts Correct
- 5 pts Wrong work of the gas
- 5 pts Wrong heat transfer of the gas
- 5 pts Wrong work of the piston
- 5 pts Wrong potential energy of the piston

QUESTION 3

3 Question 3 20 / 20

- ✓ - 0 pts Correct
- 20 pts Please solve the revised Problem 2.49 posted on the blackboard.
- 5 pts $W_{\text{paddle}} = -3$ not 3 Btu. Paddle wheel transfers energy to the gas.
- 5 pts Wrong calculation

QUESTION 4

4 Question 4 20 / 20

- ✓ - 0 pts Correct
- 5 pts Wrong shaft work
- 5 pts Wrong displacing atmosphere work
- 5 pts Wrong heat transfer to the gas
- 2.5 pts Incomplete accounting of the heat transfer of energy
- 5 pts Wrong accounting of the heat transfer of energy

QUESTION 5

5 Question 5 20 / 20

- ✓ - 0 pts Correct
- 1 pts Wrong sketch of the cycle
- 5 pts Wrong sketch of the cycle
- 5 pts Wrong heat transfer for process 1-2
- 5 pts Wrong V_2
- 5 pts Wrong W_{cycle}

Thanh Nguyen

CHE 201 - HW3

1.) Sketch

$$T_s = 47^\circ\text{C}$$
$$\varepsilon = 0.93$$



Surroundings at $T_o = 27^\circ\text{C}$
 $h = 10 \text{ W/m}^2\cdot\text{K}$

Engineering Model

1. The system is closed
2. No work is involved

Analysis

Convection: $\dot{Q}_c = hA(T_s - T_o) \rightarrow \frac{\dot{Q}_c}{A} = h(T_s - T_o)$

$$\rightarrow \frac{\dot{Q}_c}{A} = (10 \text{ W/m}^2\cdot\text{K})[(47 + 273 \text{ K}) - (27 + 273 \text{ K})] = 200 \text{ W/m}^2 \times \frac{1 \text{ kW}}{1000 \text{ W}} = 0.2 \text{ kW/m}^2$$

Radiation: $\dot{Q}_r = \varepsilon \sigma A T_s^4 \rightarrow \frac{\dot{Q}_r}{A} = \varepsilon \sigma T_s^4$

$$\rightarrow \frac{\dot{Q}_r}{A} = (0.93)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(47 + 273 \text{ K})]^4 = 705 \text{ W/m}^2 \times \frac{1 \text{ kW}}{1000 \text{ W}} = 0.705 \text{ kW/m}^2$$

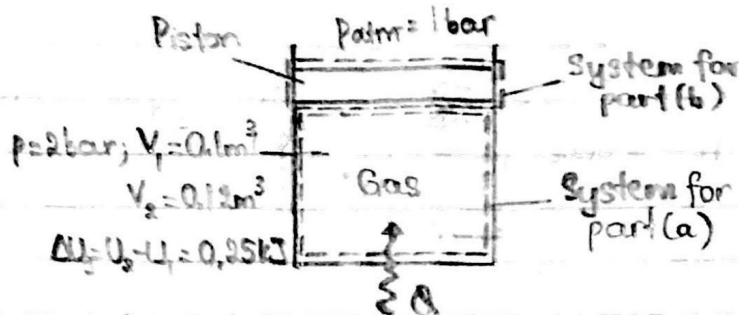
Total net rate of heat transfer

$$\frac{\dot{Q}_t}{A} = \frac{\dot{Q}_c}{A} + \frac{\dot{Q}_r}{A} = (0.2 \text{ kW/m}^2) + (0.705 \text{ kW/m}^2) = 0.905 \text{ kW/m}^2$$

1 Question 1 15 / 20

- 5 pts Wrong equation in convection
- 5 pts Wrong calculation in convection
- ✓ - 5 pts Wrong equation in radiation
- 5 pts Wrong calculation in radiation
- 0 pts correct

2.) Sketch



Engineering Model

1. Both systems (a) and (b) are closed
2. In system (a), $\Delta KE_g = \Delta PE_g = 0$
3. In system (b), $\Delta KE_p = \Delta U_p = 0$
4. No heat is lost to the cylinder walls and piston.

Analysis

(a) For the gas as the system.

Since it undergoes a constant-pressure expansion, the work is

$$W_g = \int_{V_1}^{V_2} p dV = p \Delta V = p(V_2 - V_1) = (2 \text{ bar}) \times \frac{10^5 \text{ N/m}^2}{1 \text{ bar}} \times (0.12 \text{ m}^3 - 0.1 \text{ m}^3)$$

$$= 4000 \text{ J} \times \frac{1 \text{ kJ}}{1000 \text{ J}} = 4 \text{ kJ}$$

$$\Delta KE_g + \Delta PE_g + \Delta U_g = Q - W_g \rightarrow Q = \Delta U_g + W_g = (0.25 \text{ kJ}) + 4 \text{ kJ} = 4.25 \text{ kJ}$$

(b) For the piston as the system

$$W_p = \int_{V_1}^{V_2} \sum p dV = (P_{\text{atm}} - P_{\text{gas}})(V_2 - V_1) = (1 \text{ bar} - 2 \text{ bar}) \times \frac{10^5 \text{ N/m}^2}{1 \text{ bar}} \times (0.12 \text{ m}^3 - 0.1 \text{ m}^3)$$

$$= -2000 \text{ J} \times \frac{1 \text{ kJ}}{1000 \text{ J}} = -2 \text{ kJ}$$

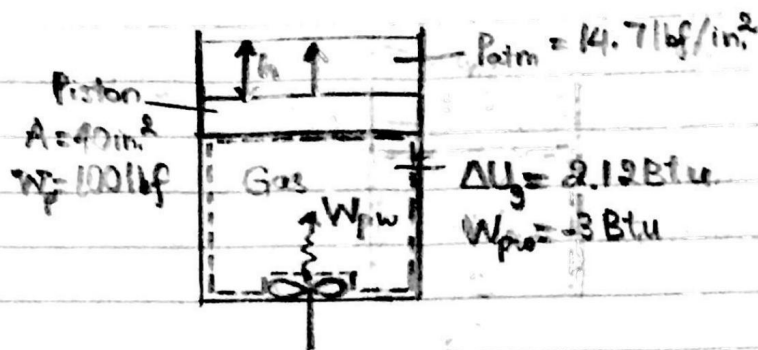
$$\Delta KE_p + \Delta PE_p + \Delta U_p = Q - W_p \rightarrow \Delta PE_p = -W_p = 2 \text{ kJ}$$

2 Question 2 20 / 20

✓ - 0 pts Correct

- 5 pts Wrong work of the gas
- 5 pts Wrong heat transfer of the gas
- 5 pts Wrong work of the piston
- 5 pts Wrong potential energy of the piston

3.) Sketch



Find h ?

Engineering Model

1. The system is closed
2. There is no heat transferred from or to the system, $Q = 0$
3. Friction can be neglected.
4. $\Delta KE_g = \Delta PE_g = 0$

Analysis

The work by the gas on the piston in order to elevate the piston with an increase of h is $W_{g \rightarrow p} = (\sum F) \cdot d = (W_p + P_{atm} A) \cdot h = [100 \text{ lbf} + (14.7 \text{ lbf/in}^2)(40 \text{ in}^2)] h$
 $= (688 \text{ lbf}) h = (688 \text{ lbf}) \times \frac{4.448 \text{ N}}{1 \text{ lbf}} \times h = (3060 \text{ N}) h$

$$\Delta U_g = (2.12 \text{ Btu}) \times \frac{1055 \text{ J}}{1 \text{ Btu}} = 2237 \text{ J}$$

$$W_{pw} = (-3 \text{ Btu}) \times \frac{1055 \text{ J}}{1 \text{ Btu}} = -3165 \text{ J}$$

$$\Delta KE_g + \Delta PE_g + \Delta U_g = Q - W$$

$$\rightarrow \Delta U_g = -W = -(W_{pw} + W_{g \rightarrow p})$$

$$\rightarrow W_{g \rightarrow p} = -\Delta U_g - W_{pw} = -2237 \text{ J} - (-3165 \text{ J}) = 928 \text{ J}$$

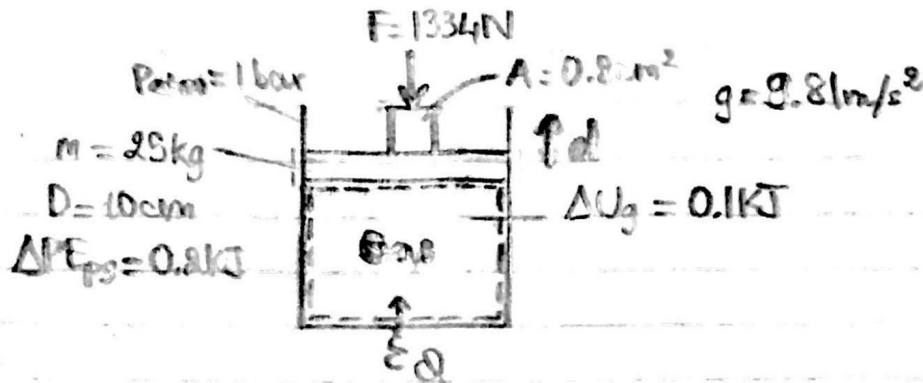
$$\rightarrow (3060 \text{ N}) h = 928 \text{ J} \rightarrow h = \frac{928 \text{ J}}{3060 \text{ N}} = 0.303 \text{ m} \times \frac{1 \text{ ft}}{0.305 \text{ m}} = 0.994 \text{ ft}$$

3 Question 3 20 / 20

✓ - 0 pts Correct

- 20 pts Please solve the revised Problem 2.49 posted on the blackboard.
- 5 pts $W_{\text{paddle}} = -3$ not 3 Btu. Paddle wheel transfers energy to the gas.
- 5 pts Wrong calculation

4) Sketch



Engineering Model

1. The system is closed
2. Friction is negligible
3. No heat is lost to the piston and cylinder
4. $g = 9.81 \text{ m/s}^2$
5. $\Delta KE_g = \Delta PE_g = 0$

Analysis

(a) $\Delta PE_{ps} = mgd \Rightarrow d = \frac{\Delta PE_{ps}}{mg} = \frac{(0.2 \text{ kJ})(1000 \text{ J})}{(25 \text{ kg})(9.81 \text{ m/s}^2)(1 \text{ kJ})} = 0.815 \text{ m}$

(a) $W_g = F \cdot d = (1334 \text{ N})(0.815 \text{ m}) = 1087 \text{ J} \times \frac{1 \text{ kJ}}{1000 \text{ J}} = 1.087 \text{ kJ}$

(b) $A_{\text{net}} = A_p - A_s = \frac{\pi D^2}{4} - A_s = \frac{\pi (0.1 \text{ m})^2}{4} - (0.8 \text{ cm}^2) \times \left(\frac{1 \text{ m}}{100 \text{ cm}}\right)^2 = 7.774 \times 10^{-3} \text{ m}^2$

$W_{\text{atm}} = P_{\text{atm}} A_{\text{net}} d = (1 \text{ bar}) \left(\frac{10^5 \text{ N/m}^2}{1 \text{ bar}}\right) (7.774 \times 10^{-3} \text{ m}^2) (0.815 \text{ m}) = 634 \text{ J}$
 $= (634 \text{ J}) \left(\frac{1 \text{ kJ}}{1000 \text{ J}}\right) = 0.634 \text{ kJ}$

(c) The work the gas done against the weight of piston-shaft is

$W_{ps} = (25 \text{ kg})(9.81 \text{ m/s}^2)(0.815 \text{ m}) = (200 \text{ J}) \times \left(\frac{1 \text{ kJ}}{1000 \text{ J}}\right) = 0.2 \text{ kJ}$

$\Delta KE_g + \Delta PE_g + \Delta U_g = Q - W \rightarrow \Delta U_g = Q - W$

$\rightarrow Q = \Delta U_g + \sum W = (0.1 \text{ kJ}) + (0.2 \text{ kJ}) + (1.087 \text{ kJ}) + (0.634 \text{ kJ}) = 2.021 \text{ kJ}$

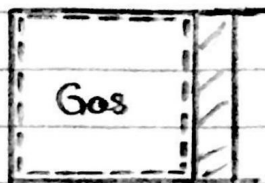
(d) When 2.021 kJ of heat is transferred to the gas, 1.087 kJ of that energy is transformed into work on the shaft, 0.634 kJ is transformed into work against the pressure of atmosphere, 0.2 kJ is transformed into work against the weight of the piston-shaft, and the remaining 0.1 kJ is preserved in the gas in the form of internal energy.

4 Question 4 20 / 20

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- 5 pts Wrong shaft work
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- 5 pts Wrong heat transfer to the gas
- 2.5 pts Incomplete accounting of the heat transfer of energy
- 5 pts Wrong accounting of the heat transfer of energy

5.) Sketch



Engineering Model

1. The system is closed
2. $\Delta KE = \Delta PE = 0$
3. Work is only in the form of volume change

Analysis

(a) $\Delta KE + \Delta PE + \Delta U = Q - W \rightarrow \Delta U = Q - W$; $W = \int_{V_1}^{V_2} p dV$

• Process 1-2: Constant pressure, $p = (1.4 \text{ bar}) \left(\frac{10^5 \text{ N/m}^2}{1 \text{ bar}} \right) = 1.4 \times 10^5 \text{ N/m}^2$

$W_{12} = \int_{V_1}^{V_2} p dV = p \Delta V = p(V_2 - V_1)$

$\rightarrow V_2 = \frac{W_{12}}{p} + V_1 = \frac{(10.5 \text{ kJ})(1000 \text{ J})}{(1.4 \times 10^5 \text{ N/m}^2)(1 \text{ kJ})} + 0.028 \text{ m}^3 = 0.103 \text{ m}^3$

• Process 3-1: constant volume $\rightarrow V_3 = V_1 = 0.028 \text{ m}^3$

• Process 2-3: $pV = \text{constant}$

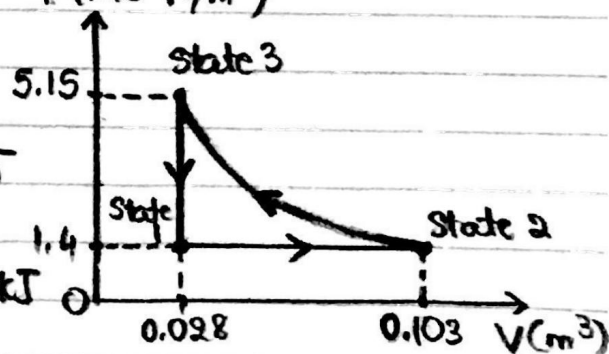
$\rightarrow p_2 V_2 = p_3 V_3 \rightarrow p_3 = \frac{p_2 V_2}{V_3} = \frac{p V_2}{V_3} = \frac{(1.4 \times 10^5 \text{ N/m}^2)(0.103 \text{ m}^3)}{0.028 \text{ m}^3} = 5.15 \times 10^5 \text{ N/m}^2$

(b) $W_{23} = \int_{V_2}^{V_3} p dV = \int_{V_2}^{V_3} \frac{p_2 V_2}{V} dV = p_2 V_2 \ln \frac{V_3}{V_2}$

$= (1.4 \times 10^5 \text{ N/m}^2)(0.103 \text{ m}^3) \left[\ln \left(\frac{0.028 \text{ m}^3}{0.103 \text{ m}^3} \right) \right] \frac{1 \text{ kJ}}{1000 \text{ J}} = -18.8 \text{ kJ}$

$W_{31} = 0$

$W_{\text{cycle}} = W_{12} + W_{23} + W_{31} = (10.5 \text{ kJ}) + (-18.8 \text{ kJ}) + 0 = -8.3 \text{ kJ}$



(c) $\Delta U_{\text{cycle}} = Q_{\text{cycle}} - W_{\text{cycle}} \rightarrow Q_{\text{cycle}} - W_{\text{cycle}} = 0 \rightarrow Q_{\text{cycle}} = W_{\text{cycle}} = -8.3 \text{ kJ}$

$U_1 - U_3 = \Delta U_{31} = -26.4 \text{ kJ} \rightarrow \Delta U_{31} = Q_{31} - W_{31} \rightarrow Q_{31} = \Delta U_{31} = -26.4 \text{ kJ}$

$U_3 = U_2 \rightarrow \Delta U_{23} = 0 \rightarrow \Delta U_{23} = Q_{23} - W_{23} \rightarrow Q_{23} = W_{23} = -18.8 \text{ kJ}$

$Q_{\text{cycle}} = Q_{12} + Q_{23} + Q_{31}$

$\rightarrow Q_{12} = Q_{\text{cycle}} - Q_{23} - Q_{31} = (-8.3 \text{ kJ}) - (-18.8 \text{ kJ}) - (-26.4 \text{ kJ}) = 36.9 \text{ kJ}$

5 Question 5 20 / 20

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- 5 pts Wrong V_2
- 5 pts Wrong W_{cycle}